

Class 10 Science - Important Topics

Physics

Light - Reflection and Refraction

Laws of Reflection

1. Angle of incidence = Angle of reflection
2. Incident ray, reflected ray, and normal lie in the same plane

Mirror Formula

$$1/f = 1/v + 1/u$$

Where f = focal length, v = image distance, u = object distance

Refraction

When light passes from one medium to another, it changes direction.

Snell's Law: $n \sin i = n \sin r$

Where n = refractive index, i = angle of incidence, r = angle of refraction

Electricity

Ohm's Law

$$V = IR$$

Where V = potential difference, I = current, R = resistance

Electrical Power

$$P = VI = I^2R = V^2/R$$

Where P = power in watts

Series Circuit

- Same current flows through all components
- Total resistance: $R = R_1 + R_2 + R_3$

Parallel Circuit

- Same voltage across all components
- Total resistance: $1/R = 1/R_1 + 1/R_2 + 1/R_3$

Magnetic Effects of Electric Current

When electric current flows through a conductor, it produces a magnetic field around it.

Fleming's Left Hand Rule

Used to find direction of force on current-carrying conductor in magnetic field.

Electromagnetic Induction

When a conductor moves in a magnetic field, an EMF is induced.

Chemistry

Chemical Reactions

Types of Chemical Reactions:

1. Combination: $A + B \rightarrow AB$
2. Decomposition: $AB \rightarrow A + B$
3. Displacement: $A + BC \rightarrow AC + B$
4. Double Displacement: $AB + CD \rightarrow AD + CB$
5. Redox: Oxidation and Reduction occur simultaneously

Acids, Bases and Salts

Acids

- Taste sour
- Turn blue litmus red
- $\text{pH} < 7$

Bases

- Taste bitter, feel soapy
- Turn red litmus blue
- $\text{pH} > 7$

pH scale: 0 to 14

pH = 7 is neutral

Metals and Non-metals

Properties of Metals:

- Good conductors of heat and electricity
- Malleable and ductile
- High melting and boiling points
- Lustre (shine)

Properties of Non-metals:

- Poor conductors
- Brittle
- Low melting and boiling points
- No lustre (except iodine)

Carbon Compounds

Carbon forms covalent bonds and shows catenation (self-linking).

Hydrocarbons:

- Alkanes: $\text{C}_n\text{H}_{2n+2}$ (single bonds)
- Alkenes: C_nH_{2n} (one double bond)
- Alkynes: $\text{C}_n\text{H}_{2n-2}$ (one triple bond)

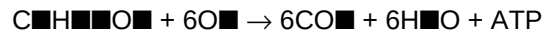
Biology

Life Processes

Seven life processes: Nutrition, Respiration, Transportation, Excretion, Control & Coordination, Growth, Reproduction

Respiration

Glucose + Oxygen $\rightarrow$ Carbon dioxide + Water + Energy



Photosynthesis

Carbon dioxide + Water → Glucose + Oxygen
(In presence of sunlight and chlorophyll)

Human Body Systems

Digestive System

Mouth → Esophagus → Stomach → Small Intestine → Large Intestine

Respiratory System

Nose → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli

Circulatory System

Heart has 4 chambers: 2 Atria and 2 Ventricles

Blood flow: Heart → Arteries → Capillaries → Veins → Heart

Heredity and Evolution

Mendel's Laws of Inheritance:

1. Law of Dominance
2. Law of Segregation
3. Law of Independent Assortment

DNA (Deoxyribonucleic Acid) carries genetic information.

Natural Selection: Organisms best adapted to environment survive and reproduce.

Study Tips

- Make concept maps
- Practice numerical problems daily
- Conduct experiments to understand concepts
- Relate science to everyday life
- Review regularly